

Branch: MCA (Data Science) Kargil	Semester: 2
Student Name: Chaitanya Sharma	UID: 25MCD10056
Subject Name: Technical Training - I Lab	Subject Code: 25CAP-652
Section/Group: 25MCD KAR-1	Date of Performance: 07-04-2026

Experiment 9

Experiment Aim:

To implement database triggers in PostgreSQL to automatically calculate values and enforce constraints during data insertion operations.

Tools used: PostgreSQL

Objectives:

- **Trigger Implementation:** To understand how to create and use triggers in PostgreSQL.
- **Automation:** To automate calculation of total payable amount using triggers.
- **Data Integrity:** To enforce constraints using trigger conditions.
- **Real-time Processing:** To execute logic automatically before inserting data.

Theory:

Triggers are special database objects that automatically execute a function when a specified event (INSERT, UPDATE, DELETE) occurs on a table. In PostgreSQL, triggers are associated with trigger functions written in PL/pgSQL. They help maintain data integrity, enforce business rules, and automate tasks.

Experiment Steps:

Step 1: Creating Table Query:

```
CREATE TABLE employee (  
  emp_id INT PRIMARY KEY,  
  emp_name VARCHAR(50),  
  working_hours INT,  
  perhour_salary NUMERIC,  
  total_payable_amount NUMERIC  
);
```

Step 2: Creating Trigger Function Query:

```
CREATE OR REPLACE FUNCTION CACULATE_PAYABLE_AMOUNT()  
RETURNS TRIGGER  
AS  
$$  
BEGIN  
  NEW.total_payable_amount := NEW.perhour_salary * NEW.working_hours;  
  
  IF NEW.total_payable_amount > 25000 THEN
```

RAISE EXCEPTION 'INVALID ENTRY BEACUSE PAYABLE AMOUNT CAN NOT BE GREATER THAN 25000';

END IF;

RETURN NEW;

END;

\$\$ LANGUAGE PLPGSQL;

Step 3: Creating Trigger

Query:

```
CREATE OR REPLACE TRIGGER  
AUTOMATED_PAYABLE_AMOUNT_CALCULATION  
BEFORE INSERT  
ON employee  
FOR EACH ROW  
EXECUTE FUNCTION CACULATE_PAYABLE_AMOUNT();
```

Step 4: Inserting Valid Data

Query:

```
INSERT INTO EMPLOYEE(emp_id, emp_name, working_hours, perhour_salary)  
VALUES (1, 'AKASH', 10, 1000);
```

Output:

```
Data Output Messages Notifications
INSERT 0 1

Query returned successfully in 1 secs 45 msec.
```

Step 5: Inserting Invalid Data (Exception Case)

Query:

```
INSERT INTO employee(emp_id, emp_name, working_hours, perhour_salary) VALUES
(2, 'Ankush', 8, 100000);
```

Output:

```
Data Output Messages Notifications
ERROR:  INVALID ENTRY BEACUSE PAYABLE AMOUNT CAN NOT BE GREATER THAN 25000
CONTEXT:  PL/pgSQL function caculate_payable_amount() line 6 at RAISE

SQL state: P0001
```

Step 6: Retrieving Data

Query:

```
SELECT * FROM EMPLOYEE;
```

Output:

Data Output Messages Notifications					
					
			SQL	Showing rows: 1 to 1 	
	emp_id [PK] integer 	emp_name character varying (50) 	working_hours integer 	perhour_salary numeric 	total_payable_amount numeric 
1	1	AKASH	10	1000	10000

Outcomes:

- I will be able to create and use triggers in PostgreSQL.
- I will understand how to automate calculations using triggers.
- I will enforce constraints using trigger conditions.
- I will understand real-time execution of database logic.